

Technical Document #831: Retrieval Augmented Generation - Comprehensive Overview

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Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Semantic search finds documents based on meaning rather than exact keywords.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Re-ranking models score retrieved documents for relevance to the query.